3 Apportionment

3.1 Hamilton's Method

lower quotas are rounded down.

Definition 3.1.1 For **Hamilton's Method**, we perform the following steps:

- Calculate the **divisor**: (total population divided by total number of representatives)
- Calculate the **quota** for each state: (state's population divided by divisor)
- Calculate **lower quotas**
- Assign any remaining representatives to those states whose decimal parts of the quota was largest.

total pop is 897,934.

Example 3.1.2 The state of Delaware has three counties: Kent, New Castle, and Sussex. The Delaware state House of Representatives has 41 members. If Delaware wants to divide this representation along county lines, let's use Hamilton's method to apportion them. The populations of the counties are as follows (from the 2010 Census): Kent: 162,310, New Castle: 538,479, and Sussex: 197,145.

Example 3.1.3 Use Hamilton's method to apportion the 75 seats of Rhode Island's House of Representatives amoung it's five counties.

County	Population
Brisol	49,875
Kent	166,158
Newport	82,888
Providence	626,667
Washington	126,979
Total	1,052,567

in order, final answer is 3, 12, 6, 45, 9.

Example 3.1.4 Three peaople invest in a treasure dive, each investing the amount listed below. The dive results in 36 gold coind. Apportion those coins to the investors.

Alice	\$7,600
Ben:	\$5,900
Calos:	\$1,400

Example 3.1.5 A state with five counties has 50 seats in their legislature. Using Hamilton's method, apportion the seats based on the 2000 census, then again using the 2010 census.

County	2000 Population	2010 Population
Jefferson	60,000	60,000
Clay	31,200	31,200
Madison	69,200	72,400
Jackson	81,600	81,600
Franklin	118,000	118,400

talk about what goes wrong here, this is called the population paradox.